

Appendix: The EXP3-IX Algorithm

A conceptual overview of Implicit Exploration for Adversarial Bandits.

A Introduction: The Bandit Problem

Imagine you are in a casino with K different slot machines, which we call **arms**. You will play for a total of T rounds (the **horizon**). At each round t , you must pick exactly one arm. Instead of random payouts, imagine a malicious adversary assigns a hidden loss $y_{t,i} \in [0, 1]$ to each machine i right before you pull. Crucially, **you only observe the loss of the machine you actually played**. You do not get to see what the other machines would have cost you. This missing information setting is known as the *Adversarial Multi-Armed Bandit* problem.

The same view applies beyond casinos: any problem with several possible actions can be written in bandit form when the method chooses one action and receives feedback only for that chosen action. In our case, an island has many mutation operators, but it can choose only one operator at a time.

Our goal is to minimize **regret**: the mathematical difference between the total loss we accumulated and the total loss we would have suffered if we had magically known the single best machine in hindsight [1].

B The Flaw in Standard EXP3

A classical algorithm to solve this is **EXP3**. Because we only see the loss for the pulled arm A_t , EXP3 estimates the hidden losses of all arms using a trick called *importance sampling*. It estimates the cumulative loss $\hat{L}_{t,i}$ as:

$$\hat{L}_{t,i} = \hat{L}_{t-1,i} + \frac{\mathbb{I}(A_t = i) \cdot y_{t,A_t}}{P_{t,i}} \quad (1)$$

Here, $P_{t,i}$ is the probability that the algorithm chose arm i , and $\mathbb{I}$ is an *indicator function* (1 if we pulled arm i , and 0 otherwise).

The problem: If we Look at the denominator We notice that if the probability $P_{t,i}$ of picking a specific arm becomes very small (e.g., 0.0001), the estimated loss shoots to infinity! This creates *massive variance*. While EXP3 guarantees good performance *on average*, in practice, this variance means any single run can perform spectacularly poorly [1].

C The Solution: EXP3-IX

To fix this variance issue, the **EXP3-IX** algorithm introduces a brilliant, tiny modification. It adds a small regularization parameter, $\gamma > 0$, to the denominator of the estimate:

$$\hat{L}_{t,i} = \hat{L}_{t-1,i} + \frac{\mathbb{I}(A_t = i) \cdot y_{t,A_t}}{P_{t,i} + \gamma} \quad (2)$$

By adding γ , the fraction is forever bounded by $1/\gamma$, entirely eliminating the variance explosion, regardless of how small $P_{t,i}$ becomes.

Why “Implicit Exploration”? As we introduced γ into the denominator we get an algorithm that underestimates the actual loss of the unpulled arms. That is, by making the unseen arms look just slightly more “cheap”, the algorithm is implicitly persuaded to explore them. So we can obtain safe exploration without the necessity of having an explicit random exploration phase!

D Algorithm Flow

The architecture of EXP3-IX forms an elegant, self-correcting feedback loop (Figure 1).

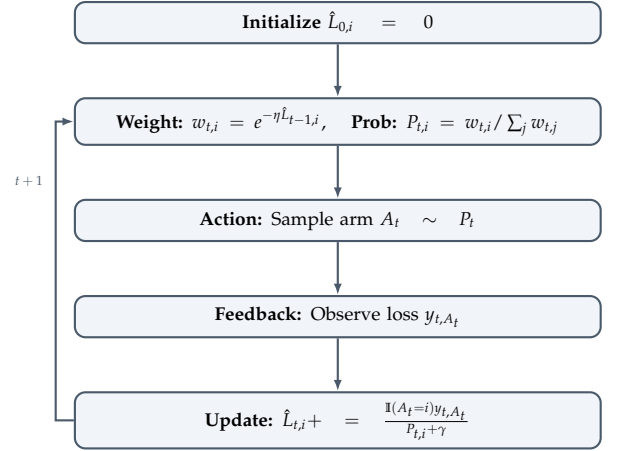


Figure 1: The continuous loop of EXP3-IX.

The step-by-step logic is detailed in Algorithm 1. The variable η (the learning rate) controls how aggressively we punish arms with high observed losses.

Algorithm 1 EXP3-IX

- 1: **Input:** Horizon T , arms K , learning rate $\eta > 0$, implicit exploration factor $\gamma > 0$.
- 2: **Initialize:** $\hat{L}_{0,i} = 0$ for all arms $i \in \{1, \dots, K\}$.
- 3: **for** $t = 1, 2, \dots, T$ **do**
- 4: $P_{t,i} = \frac{\exp(-\eta \hat{L}_{t-1,i})}{\sum_{j=1}^K \exp(-\eta \hat{L}_{t-1,j})}$.
- 5: Sample arm A_t according to P_t and observe actual loss $y_{t,A_t} \in [0, 1]$.
- 6: $\hat{L}_{t,i} = \hat{L}_{t-1,i} + \frac{\mathbb{I}(A_t = i) \cdot y_{t,A_t}}{P_{t,i} + \gamma}$ for all arms i .
- 7: **end for**

E Performance Guarantee

Because the variance is strictly controlled, EXP3-IX provides highly desirable **high-probability bounds** rather than mere expectations.

By setting the learning rate to $\eta = \sqrt{\frac{2 \log(K+1)}{TK}}$ and regularization to $\gamma = \frac{\eta}{2}$, the total regret R_T satisfies the following bound with probability exceeding $1 - \delta$ [1, 2]:

$$R_T \leq \sqrt{8TK \log(K+1)} + \mathcal{O}\left(\log\left(\frac{1}{\delta}\right)\right)$$

In plain terms: this guarantee says that, except with probability δ , the regret remains controlled even in adversarial settings. The era of catastrophic single-run failures is eliminated.

References

- [1] J. Jiao, E. Xia, and E. Liu. *Theory of Multi-Armed Bandits and RL: Lecture 12 - Exp3-Implicit Exploration*. UC Berkeley, 2020.
- [2] T. Lattimore and C. Szepesvári. *Bandit Algorithms*. Cambridge University Press, 2020.